MILITARY SPECIFICATION SHEET

ELECTRON TUBE, O-TYPE BACKWARD-WAVE

TYPE 8076A

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of MIL-E-1.

DESCRIPTION: Backward-wave oscillator, frequency range 5.43 to 5.93 GHz, 200mW, permanent-magnet focus, convection cooled, coaxial r-f connector.

Mounting position - Any (Note 5)
Weight - 25 pounds maximum
r-f Connector - Figure 1
Frequency bands - Band A - 5.43 and 5.68 GHz
Band B - 5.68 and 5.93 GHz
Power connectors - Figure 1
Cooling - Convection
Configuration - Figure 1

ABSOLUTE-MAXIMUM RATINGS: Notes 1, 2, and 3

Parameter:	Ef	Surge If	Ek	Ik	Eb	Ib	Ew	Iw	tk	Load VSWR	TA
Unit:	Vdc	A	V	mA	V	mA	V	mA	sec		°C
Maximum:	6.6	4.0	-1600	25	400	0.5	-300	10	150	(Note 4)	75
Minimum:	6.0	---	-1000	---	0	---	*0*	*0*	---	---	-55
TEST CONDITIONS:											
Test Condition (1):	6.3		NPV	NPV	NPV	---	NPV	---	---	(Note 25)	25±5
Test Condition (2):	0	---	0	---	0	---	0	---	---	---	---

GENERAL:

Qualification - Required.
Container drop - MIL-STD-1311, Method 1136 (Note 7)
Container marking - (Note 24)
Marking - (Note 6)
Voltage sequence - (Note 30)
Preparation data - (Note 8)

Method	Requirement or Test	Conditions	Symbol	Limits		Unit
				Min.	Max.	
	Qualification inspection:					
4302	*Frequency stability (1)*	*Test condition (1);*	ΔF(1)	---	±2	*Hz/MHz*
4302	Frequency stability (2)	F=5.43 and 5.93 GHz (Note 9)	ΔF(2)	---	±1	MHz
1027	Temperature cycling	Test condition (2) (Note 10)	---	---	---	---
1011	Humidity	(Note 10)	---	---	---	---

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Method	Requirement or Test	Conditions	Symbol	Limits		Unit
				Min.	Max.	
	Qualification inspection: (continued)					
1026	Temperature range	(Note 11)	---	---	---	---
---	Ballast resistor	(Note 12)	R	65	96	OHM
	Quality conformance inspection, part 1 (Note 26)					
4289	Heater current	Ef=6.3V; tk=150 sec	If	1.76	1.94	A
4250	Power output	Test condition (1);	Po	23	28	dbm
		F=5.43 to 5.93 GHz	Δ Po	---	2	db
		F=5.43 to 5.68 GHz	Δ Po	---	1	db
		F=5.68 to 5.93 GHz	Δ Po	---	1	db
	Power levels					
4250	(Variation of parameters)	Test condition (1); F=5.43 and 5.93 GHz	Po	23	28	dbm
		Δ Ef=±0.3V	Δ Po	---	1.0	db
			Δ F	---	±2.5	MHz
		Δ Eb=±1% (Notes 14 and 27)	Δ Po	---	0.5	db
---	Frequency vs. voltage	F=5.43 to 5.68 GHz	Δ Ew	---	190	V
		Test condition (1)	Eb	100	300	V
			Ek	-1000	-1600	V
		F=5.68 to 5.93 GHz	Δ Ew	---	190	V
			Eb	100	300	V
			Ek	-1000	-1600	V
		F=5.43 to 5.93 GHz (Notes 3, 15)	Δ F/Δ Ew	36/30	50/30	MHz/V
---	Transient frequency response	Test condition (1); F=5.43, 5.655, 5.705, 5.93 GHz (Note 13)	---	---	---	---
	Quality conformance inspection, part 2 (Notes 16 and 17)					
---	Vibration	Test condition (1); F=5.93 GHz (Note 18)	Δ F	---	50	KHz
1042	Shock	Test condition (2); (Note 19)	---	---	---	---
---	Spectral power output	Test condition (1); F=5.43 and 5.93 GHz (Note 20)	---	---	---	---

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Method	Requirements or Test	Conditions	Symbol	Limits		Unit
				Min.	Max.	
	Quality conformance inspection, part 2 (continued) (Notes 16 and 17)					
---	Noise power output	Test condition (1); F=5.43 and 5.93 GHz (Note 28)	Pn	---	38	db/KTB
---	Dimensions	Figure 1	---	---	---	---
	Quality conformance inspection, part 3 (Notes 17 and 21)					
	Life-test provisions	Test condition (1); F=5.93 GHz, Group S (Notes 22, 29, and 31)	t	2500	---	hours
			Ib	---	0.5	ma
	Life-test end points	Test condition (1); F=5.93 GHz (Note 23)	Po	22	---	dbm

NOTES:

1. Typical operating conditions for this device are described by Test Condition (1). Ratings shall refer to each individual rating with the tube otherwise operating in the above test condition. Note that ratings are not of themselves test conditions or operating conditions. The maximum and minimum ratings are limiting values beyond which the serviceability of any individual tube may be impaired.
2. Helix and cathode voltage are referenced to the shell which normally is at chassis potential. Anode voltage is referenced to the cathode. The collector is connected internally to the shell. The positive leg of the heater shall be connected externally to the cathode during these tests.
3. The symbol NPV used herein refers to the values shown on the nameplate in accordance with note 6. Ek, Ew, and Eb are the respective cathode, helix, and anode voltages which produce power output and frequency within the specified limits.
4. The tube shall be capable of operating into a load of infinite VSWR without damage to the tube.
5. A 2-inch clearance is required between the tube external surfaces and adjacent unmagnetized magnetic material. Permanent changes in power and frequency shall be less than ±0.5db and ±1.0 MHz, respectively, when unmagnetized magnetic material is placed within 2 inches of the external surfaces of the tube, and removed.
6. A nameplate, located as shown in Figure 1, shall be affixed to the tube and shall be marked in accordance with MIL-E-1, and as follows:

Electrode-voltage table

Parameter	Band A	Band B	Unit
Ek			V
Ik			mA
Eb			V
Ew(low)			V
Ew(high)			V

7. The tube thereafter shall meet the requirements of quality conformance inspection, part 1.
8. Two prints of data sheets for all tests performed under this specification shall be supplied to the procuring activity. One copy of the data shall be packed with the tube.

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NOTES: (continued)

9. Apply heater voltage, and 150 seconds thereafter, at t_o, apply all beam voltages. Start timing upon application of heater voltage. Then, t_o=150 seconds, t_1=300 seconds, t_2=360 seconds. Frequency stability (1) shall be measured in the time between t_1 and t_2 during an interval not exceeding 0.5 seconds. Frequency stability (2) shall be the frequency change between t_1 and t_2.

10. The temperature shall be reduced to -55° C and then shall be increased to 75° C. The tube then shall be subjected to the specified humidity test; the temperature range shall be 70° C to 80° C. After these tests the tube shall meet the requirements for power output under quality conformance inspection, part 1.

11. The tube shall be operated at temperatures of 0° C and 50° C. The tube shall be maintained in the non-operating condition for three hours at each of the temperatures. After cathode warmup for time tk, at the end of each three-hour period, the tube shall be operated and shall meet the requirements for power output under quality conformance inspection, part 1.

12. The resistor shall consist of two 40-ohm resistors connected in series. These resistors shall be Dale Manufacturing Company Model RH50, or equivalent.

13. With the tube operating normally at the specified frequency, apply a superimposed pulse on the helix voltage of such amplitude as to produce a frequency change of 225 MHz in each direction. The rise time of the pulse shall be not greater than 1.0 milliseconds. The total frequency change in the time interval 2.83 milliseconds to 4.63 milliseconds after the start of the pulse shall be not greater than 10KHz. The total change in helix voltage during the above interval shall not exceed 1.5 mV peak to peak.

14. When Ef and Eb are varied as specified, with all other voltages remaining constant, the power output, the total power output variation, and the frequency shall be within the specified limits.

15. The frequency/helix-voltage characteristic shall be a monotonically increasing function, the slope of which is within the specified limits in all 30 MHz intervals over the specified frequency range.

16. Quality conformance tests, part 2, shall be performed in accordance with MIL-STD-105, inspection level II, with an AQL of 4 percent. In the event of lot rejection, tightened inspection shall be invoked. Normal inspection shall be resumed when two consecutive lots have conformed with the tests of quality conformance inspection, part 2. For inspection purposes the lot size shall be one month's production.

17. Tubes for these tests shall be selected at random from the production lot, and shall have passed the tests for quality conformance inspection, part 1.

18. The tube shall be vibrated in the operating condition in accordance with MIL-STD-167(SHIPS), type 1. The variation of output frequency attributable to vibration shall not exceed the specified limit.

19. Use the sawtooth shock pulse with peak acceleration of 40g, and duration of 10 m sec. The tube thereafter shall be tested for quality conformance inspection, part 1, large-signal power output, and shall operate within the specified limits.

20. The sideband ratio relative to the carrier shall be not greater than that shown in the following table:

Frequency, MHz	Power, ratio db
F+0.9375	-49
F-0.9375	-49
F+1.875	-61
F-1.875	-61
F+3.750	-73
F-3.750	-73
F+7.500	-85
F-7.500	-85
F+15.000	-97
F-15.000	-97
F+30.000	-109
F-30.000	-109
F+60.000	-97
F-60.000	-97

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The sidebands at 30 MHz points thereafter, to F=5.43 and F=5.93 MHz shall not increase at a rate greater than 12 db for each octave of 30 MHz and shall in no event display a power ratio of more than -49 db.

21. Tubes selected for this test shall previously have passed the tests for shock and vibration.

22. The tube shall be operated intermittently according to the following schedule for the duration of the life test:

    (a) Apply Ef for 150 seconds
    (b) Apply all other voltages
    (c) Operate for $7\frac{1}{2}$ hours
    (d) Remove all voltages for remainder of an 8-hour cycle
    (e) Repeat (a) through (e)

23. End of life shall be that point at which the power output has fallen to 22 dbm, or the tube will not pass the test for transient frequency response. These criteria shall apply at any frequency in either band.

24. The container shall be marked with the following legend: "Magnetic materials. Store in original container until ready for installation."

25. All requirements specified herein shall be met with any load VSWR of 1.25:1 or less, at any phase of the load.

26. Inspection shall be 100 percent.

27. The minimum and maximum values of Po under Variation of parameters shall include changes due to the specified variations of parameters.

28. The noise power output measured at 30 MHz at either side of the carrier shall not exceed the specified value.

29. The lot size shall be one month's production.

30. The tube voltages shall be applied in the following sequence:

    (a) Heater voltage (allow 150 seconds warmup)
    (b) Helix voltage
    (c) Cathode voltage
    (d) Anode voltage

    The voltages shall be removed in the reverse sequence. Helix, cathode and anode voltage may be applied or removed simultaneously.

31. Following storage for a period of one year, the tube shall be capable of passing all of the requirements of this specification, and shall require no special burn-in procedure. If a burn-in procedure is required for tubes stored for longer periods, such procedure shall be specified by the contractor.

User activities:
Navy-AS,OS,MC,CG,SH

Preparing activity:
Navy - EC

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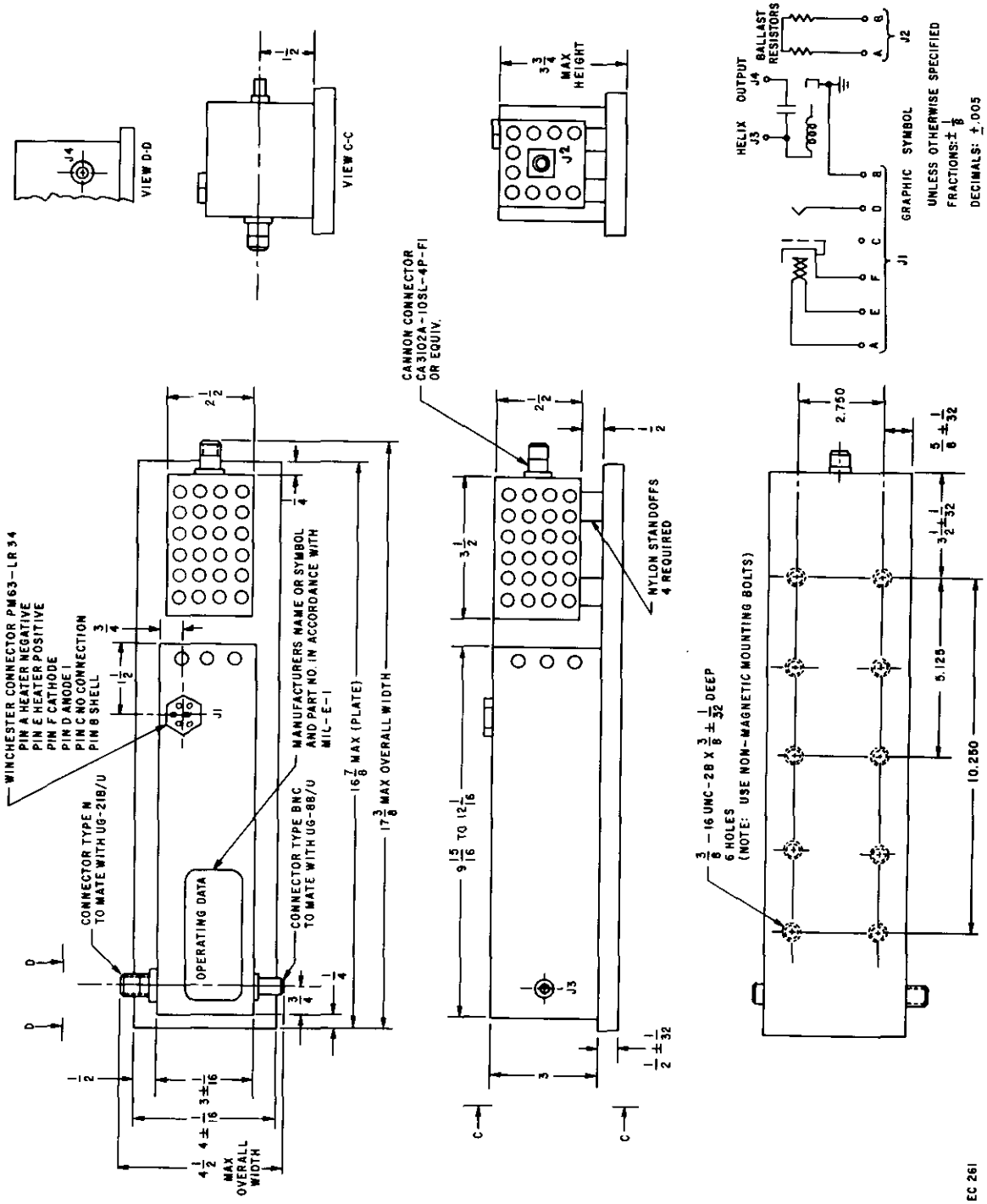


Figure 1 - Outline drawing of electron tube type 8076A

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